

Gourav Lohar

AI / Machine Learning Engineer

Panagarh, West Bengal

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EXPERIENCE

EBest Solutions Pvt. Ltd.

Software Engineer – AI / ML

Dec 2025 – Present

Durgapur, West Bengal

- Engineered and deployed a production-grade, self-hosted **AI Voice Agent Platform** on **AWS** using **Docker**, powering **4,000+ AI voice interactions**. Designed a cost-efficient **cascaded STT–LLM–TTS architecture** as an alternative to expensive real-time multimodal models, reducing inference costs while maintaining conversational quality. Co-located speech and language models within the same infrastructure and implemented **Voice Activity Detection (VAD)** and **End-of-Utterance (EOU) detection** to reduce end-to-end response latency, along with **RAG**, **persistent agent memory**, **tool calling**, and **Plivo APIs**.
- Deployed and operated production voice agents for a **London-based restaurant**, **Serbian restaurant**, and **Serbian dental clinic**, supporting automated ordering, customer assistance, and appointment workflows. Built telephony infrastructure using **Asterisk**, **SIP trunking**, and **private VMs** for Serbian calling and integrated an **Indian BSNL landline** into the AI voice-agent infrastructure.
- Developed a high-throughput **OpenCV-based OMR processing and evaluation system** that processed **700,000+ university examination sheets** across multiple universities and diverse OMR formats. Automated **student roll-number extraction**, **QR-based answer-sheet identification**, **examination metadata extraction**, and **set-wise MCQ evaluation**, achieving over **98.6% automated accuracy** and significantly reducing manual verification.
- Designed and developed an **Intelligent Document Processing (IDP) platform** for automated extraction and structuring of information from **invoices**, **legal documents**, **passports**, and **other complex document types**. Integrated open-source **Vision-Language Models (VLMs)**, **OCR models**, and **MinerU** for layout-aware document understanding, with **GPU-accelerated inference using vLLM** to improve extraction throughput and support scalable document-processing workflows.
- Developed an **agentic hospital appointment chatbot** using **LangGraph**, implementing **intent routing**, **slot filling**, **tool calling**, and **stateful workflows**. Integrated persistent multi-turn conversation memory using a **PostgreSQL checkpoint** to preserve agent state and contextual continuity across interactions.

Capsitech Pvt. Ltd.

Assistant ML Engineer – L1

Jun 2025 – Nov 2025

Jodhpur, Rajasthan

- Engineered a production-oriented **document intelligence and OCR pipeline** using **FastAPI** and **PaddleOCR**, achieving over **95% extraction accuracy** on complex HR documents and transforming unstructured content into structured, machine-readable information.
- Implemented a **multi-threaded job-processing architecture** integrating **Ollama LLM (Gemma 3 4B)** for intelligent information extraction, enabling concurrent **OCR and LLM inference** and reducing processing time by approximately **60%** compared with the previous pipeline.
- Fine-tuned **HuggingFaceTB/SmolLM2-360M-Instruct** on a domain-specific dataset to improve structured information extraction and context-aware text generation for document-processing workflows.

PROJECTS

BioLens | TensorFlow, Flask, Gemini API, Python

2025

- Built an AI-powered **crop disease diagnosis platform** using a **TensorFlow deep learning model** trained on **87,000+ crop leaf images** spanning **38 disease classes**, enabling automated disease identification from uploaded crop images.
- Integrated an **LLM-powered agricultural chatbot** and intelligent crop scheduling assistant using the **Gemini API** to provide contextual farming recommendations, disease-management guidance, and crop cultivation planning.

Research Agent | Python, Flask, LangChain, Qwen QwQ-32B, Azure

2025

- Developed an **AI-powered research agent** using **LangChain** and **Qwen QwQ-32B** for advanced query understanding, information retrieval, and multi-source knowledge synthesis across research-oriented queries.
- Implemented a **zero-shot agentic routing architecture** with tool calling to dynamically select between **ArXiv** and **Wikipedia APIs** based on query intent, enabling source-aware information retrieval and response synthesis without task-specific training.

TECHNICAL SKILLS

Languages: Python, SQL

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, OpenCV, PaddleOCR, NumPy, RAG, Agentic AI, vLLM, LLM Fine-Tuning, Prompt Engineering

Backend & Tools: FastAPI, Flask, PostgreSQL, MinIO, AWS, Docker, Linux, Git, GitHub, Bitbucket, NGINX

EDUCATION

NSHM Institute of Engineering and Technology

2021 – 2025

B.Tech. in Artificial Intelligence and Machine Learning – **CGPA: 8.97**

Durgapur, West Bengal

Nepali Para Hindi High School

2020 – 2021

Higher Secondary – **76%**

Durgapur, West Bengal